

# Multiplicative Inverse, Fermat's little Theorem

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# Overview

- 1 Multiplicative Inverses (8.6.1)
- 2 Cancellation (8.6.2)
- 3 Fermat's Little Theorem (8.6.3)

## Back to basics

Definition: The *multiplicative inverse* of a number  $x$  is a number  $x^{-1}$  such that:  
 $x \cdot x^{-1} = 1$ .

Division by  $x$  is really multiplication by  $x^{-1}$ .

Over the reals, what values have inverses? Everybody but zero.

Over the integers, what values have inverses? Only 1 and  $-1$ .

Over the integers mod  $n$ , what values have inverses?

## Example, mod 10

|   | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 2 | 0 | 2 | 4 | 6 | 8 | 0 | 2 | 4 | 6 | 8 |
| 3 | 0 | 3 | 6 | 9 | 2 | 5 | 8 | 1 | 4 | 7 |
| 4 | 0 | 4 | 8 | 2 | 6 | 0 | 4 | 8 | 2 | 6 |
| 5 | 0 | 5 | 0 | 5 | 0 | 5 | 0 | 5 | 0 | 5 |
| 6 | 0 | 6 | 2 | 8 | 4 | 0 | 6 | 2 | 8 | 4 |
| 7 | 0 | 7 | 4 | 1 | 8 | 5 | 2 | 9 | 6 | 3 |
| 8 | 0 | 8 | 6 | 4 | 2 | 0 | 8 | 6 | 4 | 2 |
| 9 | 0 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |

What specific values have inverses? 1, 3, 7, 9.

What specific values do *not* have inverses? 0, 2, 4, 5, 6, 8.

General rule?  $a$  has an inverse (mod  $n$ ) iff  $\gcd(a, n) = 1$ .

## Back to solving

$$3x + 4 \equiv 27 \pmod{11}$$

$$3x \equiv 23 \pmod{11} \quad \text{add } -4 \text{ to both sides}$$

Want to multiply both sides by  $3^{-1} = 4$ , since  $3 \times 4 \equiv 1 \pmod{11}$ .

$$3x \equiv 23 \pmod{11}$$

$$4 \times 3x \equiv 4 \times 23 \pmod{11} \quad \text{multiply both sides by 4}$$

$$12x \equiv 92 \pmod{11} \quad \text{simplify}$$

$$x \equiv 4 \pmod{11} \quad \text{congruence}$$

Double check: 16 vs. 5, 11 divides the difference!

## Inverse mod prime

General rule for existence of multiplicative inverses? Perhaps  $a$  has an inverse iff  $\gcd(a, n) = 1$ ?

If this rule holds, all values (except zero!) have inverses mod a prime.

**Lemma:** If  $p$  is prime and  $k$  is not a multiple of  $p$ , then  $k$  has a multiplicative inverse modulo  $p$ .

**Proof:** Since  $p$  is prime and  $k$  is not a multiple of  $p$ ,  $\gcd(p, k) = 1$ . Therefore, there are  $s$  and  $t$  such that  $1 = sp + tk$ . So, mod  $p$ , that's  $1 \equiv tk$ , or  $t \equiv k^{-1} \pmod{p}$ . QED.

Example: What's the multiplicative inverse of 3 (mod 11)?

$$\text{gcdcombo}(3, 11) = (4, -1, 1)$$

So? 4 works. Because  $1 = 4 \times 3 - 1 \cdot 11$ , so, mod 11, that's  $1 = 4 \times 3$ .

## Back to dividing both sides

Earlier, we saw:

$$7 \equiv 28 \pmod{3}$$

$$1 \equiv 4 \pmod{3} \quad \text{divide by 7}$$

Doesn't actually work, in general:

$$12 \equiv 6 \pmod{3}$$

$$4 \not\equiv 2 \pmod{3} \quad \text{divide by 3}$$

Why? Because we're really talking about multiplying both sides by  $0^{-1}$ , which doesn't exist.

Apart from dividing by 0, we can cancel.

## Cancellation proof

If we have

$$ak \equiv bk \pmod{p}$$

and  $p$  is prime and  $k \not\equiv 0 \pmod{p}$ , then  $k^{-1} \pmod{p}$  exists.

Multiply both sides by  $k^{-1}$  and congruence is maintained.



## Never need to multiply big numbers

When doing multiplication mod  $n$ , we can always mod  $n$  the numbers first.

Example:

$$\begin{aligned} 7415 \times 2993 \bmod 3 \\ = 22193095 \bmod 3 \\ = 1 \end{aligned}$$

OR:

$$\begin{aligned} (7415 \bmod 3) \times (2993 \bmod 3) \bmod 3 \\ (2 \times 2) \bmod 3 \\ = 1. \end{aligned}$$

# Proof

$$ab \bmod n = (a \bmod n)(b \bmod n) \bmod n.$$

$$a = q_1n + r_1$$

$$b = q_2n + r_2$$

$$ab = (q_1n + r_1)(q_2n + r_2)$$

$$ab = (q_1q_2n + q_1r_2 + q_2r_1)n + r_1r_2$$

## Permuting

**Corollary:** Suppose  $p$  is prime and  $k$  is not a multiple of  $p$ . Then, the sequence of remainders on division by  $p$  of the sequence:

$$1 \cdot k, 2 \cdot k, \dots, (p-1) \cdot k$$

is a permutation of the sequence:

$$1, 2, \dots, (p-1).$$

Example,  $k = 3, p = 11$ :

|                 |   |   |   |    |    |    |    |    |    |    |
|-----------------|---|---|---|----|----|----|----|----|----|----|
| $i$             | 1 | 2 | 3 | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
| $\times k$      | 3 | 6 | 9 | 12 | 15 | 18 | 21 | 24 | 27 | 30 |
| $\text{mod } p$ | 3 | 6 | 9 | 1  | 4  | 7  | 10 | 2  | 5  | 8  |

## Permutation proof

**Proof:** The sequence of remainders contains  $p - 1$  numbers. Since  $i \times k$  is not divisible by  $p$  (neither contains a factor of  $p$ ) for  $i = 1, \dots, p - 1$ , all these remainders are in  $[1, p)$  by the definition of remainder.

Claim: if  $i \cdot k \equiv j \cdot k \pmod{p}$ , then  $i = j$ . (Cancel  $k$ ; since  $1 \leq i < p$ ,  $i \pmod{p} = i$ , same for  $j$ .)

So,  $i - 1$  distinct values between 1 and  $i - 1$ : it's a permutation.

It's a magic shuffle function. Useful for randomization and sending secret messages!

## Fermat's little theorem

**Theorem:** Suppose  $p$  is prime and  $k$  is not a multiple of  $p$ . Then:

$$k^{p-1} \equiv 1 \pmod{p}.$$

$$(p-1)!$$

$$= 1 \cdot 2 \cdot \dots \cdot (p-1)$$

|               |                                                                           |                             |
|---------------|---------------------------------------------------------------------------|-----------------------------|
| <b>Proof:</b> | $= \text{rem}(k, p) \cdot \text{rem}(2k, p) \cdots \text{rem}((p-1)k, p)$ | Defn. of factorial          |
|               | $\equiv k \cdot 2k \cdots (p-1)k \pmod{p}$                                | Permutation lemma           |
|               | $\equiv (p-1)!k^{p-1} \pmod{p}$                                           | Congruence of mult. algebra |

Note that  $(p-1)!$  is not a multiple of  $p$  because none of  $1, 2, \dots, (p-1)$  contain a factor of  $p$ . So, by the Cancellation lemma, we can cancel  $(p-1)!$  from the top and bottom, proving the claim. QED

## Inverses from Fermat's little theorem

Since  $k^{p-1} \equiv 1 \pmod{p}$  and  $k^{p-1} = k \cdot k^{p-2}$ , that tells us that  $k^{p-2}$  is the multiplicative inverse for  $k$ .

We can compute  $k^{p-2} \pmod{p}$  efficiently using a technique called exponentiation by repeated squaring.

Running time is  $2 \log p$ , just like “gcdcombo”.

## Exponentiation by Repeated Squaring Idea

Can always compute  $a^k$  by  $k - 1$  multiplications of  $a$ .

If  $k$  is even, can compute it with  $k/2 - 1$  multiplications of  $a$  to get  $a^{k/2}$ . Then,  $a^k = (a^{k/2})^2$ . So, one more multiplication and we're there.

If  $k$  is odd, similar trick to get  $a^{(k-1)/2}$ , then square, then multiply one more  $a$ .

Repeating this idea, the number of multiplications is on the order of  $2 \log k$ .

## Exponentiation by Repeated Squaring

```
def repsq(a,k):  
    if k == 0: return(1)  
    if k % 2 == 0:  
        sqroot = repsq(a,k/2)  
        return(sqroot*sqroot)  
    sqrootdiva = repsq(a,(k-1)/2)  
    return(sqrootdiva*sqrootdiva*a)
```



## Exponentiation by Repeated Squaring Mod Style

```
def repsqmodn(a,k,n):  
    a := a % n  
    if k == 0: return(1)  
    if k % 2 == 0:  
        sqrootdiva = repsqmodn(a,k/2,n)  
        return((sqrootdiva*sqrootdiva) % n)  
    sqrootdiva = repsqmodn(a,(k-1)/2,n)  
    return((sqrootdiva*sqrootdiva*a) % n)
```